# Travelling Salesman Problem

## PROGRAM:

```
from itertools import permutations

dist = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

def calculate_cost(path, dist):
    cost = 0
    for i in range(len(path) - 1):
        cost += dist[path[i]][path[i + 1]]
    cost += dist[path[-1]][path[0]]
    return cost

def tsp_brute_force(dist):
    n = len(dist)
    cities = list(range(1, n))

    min_cost = float('inf')
    best_path = None

    for perm in permutations(cities):
        path = [0] + list(perm)
        cost = calculate_cost(path, dist)
        if cost < min_cost:
            min_cost = cost
            best_path = path

    best_path.append(0)
    return tuple(best_path), min_cost

path, cost = tsp_brute_force(dist)
print(f"Path: {path}")
print(f"Minimum Cost: {cost}")
```

**OUTPUT:**

```
IDLE Shell 3.13.12
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
======================= RESTART: D:\New folder (9)\tsp.py ======================
Path: (0, 1, 3, 2, 0)
Minimum Cost: 80
>>>
```